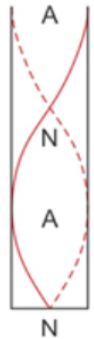


- 4 (a) (i) Fundamental mode of vibration, $\lambda = 4L = 0.680$ or $L = 0.17$ m (not formed)

1st overtone: $\lambda = \frac{4L}{3} = 0.680$ or $L = 0.51$ m (formed) **A1**

2nd overtone: $\lambda = \frac{4L}{5} = 0.680$ or $L = 0.85$ m (formed) **A1**

(ii)



A at open end, N at closed end;
An A and N in between, equally
spaced (by eye)

A1

- (b) waves pass through/enter the slits in the grating **B1**

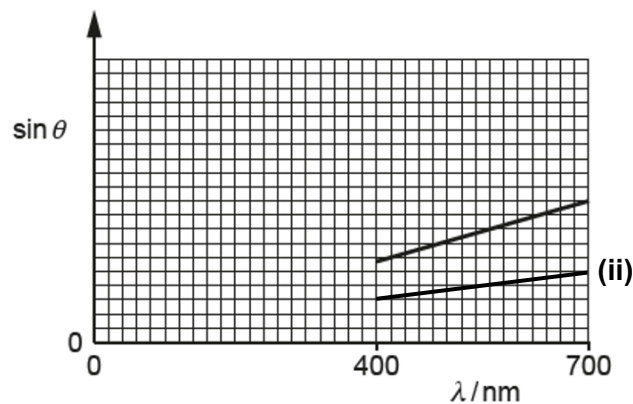
waves spread out after passing through/entering the slits **B1**

- (c) (i) $d \sin \theta = n\lambda$ **C1**

or $\sin \theta = \frac{n}{d} \lambda$

Hence $G = \frac{n}{d}$ or $d = \frac{4}{G}$ **A1**

(ii)



straight line from 400 nm to 700 nm that is always below given line

M1

straight line has smaller gradient than given line and is 5 small squares
high at wavelength 700 nm **A1**

5 (c) The velocity v may be resolved into two components – one parallel to the